RIRPA in Vayesta

Current state, cost, and a plan for HHT/CDERI improvements

10 November 2026

Nothing in this note was run. All scaling statements are operation counts. The measured numbers in Section 3.1 are quoted from the `monomial_GW` and `Davidson-gw` records, and the source is named each time.

Symbol	Meaning
o, v	occupied / virtual orbitals of the full system
ov	number of spatial particle–hole pairs, $o\,v$ (spin-orbital codes use $2ov$)
N_{aux}	number of auxiliary (density-fitting) functions; $\mathcal{O}(N)$
L	occupied–virtual CDERIs, shape (N_{aux}, ov)
D	diagonal of orbital-energy differences $\epsilon_a - \epsilon_i$, length ov
n_t	number of rows of `target_rot`, the space the moment is projected onto
N_p	number of quadrature points (Clenshaw–Curtis) or poles (HHT)
N_{frag}	number of fragments in an embedding calculation

1 Vayesta as a whole

Vayesta is a Python quantum-embedding package built on PySCF. It partitions a molecule or solid into fragments, builds a cluster for each fragment (fragment orbitals plus a bath), solves each cluster with a correlated solver (CCSD, FCI, MP2, EBFCI, ...), and assembles global properties from the fragment results. The main subpackages are:

- `core/`: fragmentation, bath construction (`core/bath/`: DMET, MP2 and RPA bath natural orbitals, bosonic baths), integrals (`core/eris.py`, where `get_cderi` lives), screening (`core/screening/`), and the generic embedding driver `core/qemb/`.
- `ewf/`, `dmet/`, `edmet/`: the embedding methods themselves (wave-function embedding, DMET, and extended/bosonic DMET).
- `solver/`: cluster solvers, including the electron–boson ones that EDMET needs.
- `rpa/`: full-system RPA. `rpa.py`, `ssrpa.py` and `ssurpa.py` are dense $\mathcal{O}(N^6)$ reference implementations. `rirpa/` is the density-fitted, moment-based code that the embedding layers use in production.

RIRPA matters because it supplies *global long-range information to local clusters*. It is called from four places:

Caller	What it asks RIRPA for	`target_rot`
`core/screening/screening_moment.py` (`mrpa` screening, cRPA)	zeroth dd moment, then only $T_f\,\eta_0\,T_f^{\mathsf{T}}$ per fragment	stacked cluster ph spaces, spin-orbital, $n_t = \sum_f ov_f^{\mathrm{clus}}$
`core/bath/rpa.py` (RPA bath NOs)	zeroth moment, then $m_0 T^{\mathsf{T}}$	$R_o \otimes R_v$: $o\,v_{\mathrm{clus}}$ or $o_{\mathrm{clus}}\,v$ rows; **one RIRPA object per fragment**
`core/qemb/qemb.py` (bosonic bath; external dRPA correction)	zeroth moment / total energy	bosonic targets; or L itself (cumulant energy)
`edmet/edmet.py`	zeroth moment and energy **with an xc kernel** (`rixc`)	cluster ph and bosonic spaces

2 How `vayesta/rpa/rirpa` is implemented

2.1 Files and classes

File	Contents
`__init__.py`	`ssRIRPA(mf, ...)` factory: UHF goes to `ssRIURPA`, a non-`None` `rixc` goes to `ssRIRRPA`, and everything else goes to `ssRIdRRPA`.
`RIRPA.py`	`ssRIRRPA`: the general (xc-kernel-capable) restricted class, which works in the **spin-orbital** $2ov$ space; also `construct_product_RI`, `construct_inverse_RI` and `compress_low_rank`.
`RIdRRPA.py`	`ssRIdRRPA`: specialised direct-RPA RHF path in **spatial** ov with explicit spin factors (the path most embedding calls take).
`RIURPA.py`	`ssRIURPA`: unrestricted variant; overrides only the integral/xc construction.
`NI_eval.py`	Numerical-integration framework: Clenshaw–Curtis on infinite and semi-infinite ranges with a nested error estimate, Gauss–Laguerre, and the optimisation of the grid scale a.
`momzero_NI.py`	Integrands for the zeroth moment (`MomzeroDeductNone/D/HigherOrder`, plus the `_dRHF` variant) and the analytic-offset integrand (`MomzeroOffsetCalcGaussLag`).
`energy_NI.py`	Integrand for $\mathrm{Tr}(MP)^{1/2}$ (`NITrRootMP`, plus `_dRHF`).

2.2 The mathematics as coded

This follows Furche (PRB **64**, 195120) and arXiv:2301.09107. For restricted dRPA in spatial orbitals,

$$A - B = D \quad \text{(diagonal)}, \qquad A + B = D + 4L^{\mathsf{T}}L, \tag{1}$$

$$MP \equiv (A-B)(A+B) = D^2 + S_L^{\mathsf{T}} S_R \quad (\text{rank } N_{\mathrm{aux}}), \qquad \eta_0 = (MP)^{1/2}(A+B)^{-1}. \tag{2}$$

Vayesta stores $A + B$ as $(\sqrt{2}L)^{\mathsf{T}}(\sqrt{2}L)$ with a further factor of 2 for spin. The square root is evaluated as a frequency integral,

$$(MP)^{1/2} = \frac{1}{\pi}\int_{-\infty}^{\infty} \Big[\, I - \omega^2 (MP + \omega^2)^{-1} \Big]\, d\omega, \tag{3}$$

and Woodbury reduces the resolvent to auxiliary space. At each frequency,

$$F(\omega) = (D^2 + \omega^2)^{-1} \quad \text{(diagonal)}, \tag{4}$$

$$Q(\omega) = S_R\, F(\omega)\, S_L^{\mathsf{T}} \quad (N_{\mathrm{aux}} \times N_{\mathrm{aux}}) \qquad \leftarrow \text{ the } N_{\mathrm{aux}}^2 ov \text{ step}, \tag{5}$$

$$\mathrm{term}(\omega) = (TF)\, S_L^{\mathsf{T}} \big[(I+Q)^{-1} - I\big]\, S_R F \quad (n_t \times ov). \tag{6}$$

The default `"HO"` scheme subtracts the diagonal and the first-order term $-I$ inside the integrand and adds them back analytically. The first-order offset,

$$T\int_0^\infty e^{-\omega D}\, S_L^\mathsf{T} S_R\, e^{-\omega D}\, d\omega, \qquad \text{exactly} \quad \frac{(S_L^\mathsf{T} S_R)_{pq}}{D_p + D_q}, \tag{7}$$

would cost $\mathcal{O}(n_t\, ov^2)$ if done exactly. Vayesta therefore evaluates it with a **second, separate 48-point Gauss–Laguerre integral**, whose scale is also optimised.

The remaining steps of the zeroth moment are as follows.

1. Fit the grid scale a for each integral (`opt_quadrature_diag`). The fit uses an exponential search over 14 decades, then Newton or bounded minimisation, so that the rule integrates a diagonal approximation exactly. Each probe costs $\mathcal{O}(N_p\, ov)$.
2. Integrate with Clenshaw–Curtis (`npoints=48`, rounded up to a multiple of 4). The $\frac{1}{2}$- and $\frac{1}{4}$-grids are nested inside it, and they give the error estimate.
3. Form the explicit inverse `np.linalg.inv(I + Q)` at every point.
4. Multiply by $(A+B)^{-1}$, itself built as an RI correction (`construct_inverse_RI`: an $N_\text{aux} \times N_\text{aux}$ inverse plus an SVD; or, in the dRPA path, an explicit $\left(I + 2LD^{-1}L^\mathsf{T}\right)^{-1}$).
5. Optionally compress (`compress > 0`) by a non-symmetric `eig` of $S_L S_R^\mathsf{T}$ with an absolute 10^{-12} cut. This is off by default.

Higher moments use the exact recurrence $\eta_{n+2} = \eta_n D^2 + (\eta_n S_R^\mathsf{T}) S_L$. Each order is two GEMMs of $\mathcal{O}(n_t N_\text{aux} ov)$, which is already optimal.

The energy (`kernel_energy`) is $E_c = \frac{1}{2}\big[\operatorname{Tr}(MP)^{1/2} - \operatorname{Tr} A\big]$. The trace is another optimised Clenshaw–Curtis integral (`NITrRootMP`), and its diagonal and first-order parts are added analytically.

The xc-kernel path (`ssRIRRPA` with `rixc`, used by EDMET) differs in two ways. First, $A - B$ is no longer diagonal, so `construct_product_RI` builds the non-symmetric product with rank $N_\text{aux}(A+B) + N_\text{aux}(A-B)$, which is up to 2–3× the CDERI rank. Second, the whole calculation runs in the $2ov$ spin-orbital space.

`get_gap` runs a **non-symmetric** Davidson (`pyscf.lib.eig`, left and right vectors) on MP.

2.3 Current cost

Per call of `kernel_moms(0, target_rot)` on the dRPA RHF path (`ssRIdRRPA`), with $N_p = 48$:

Step	Cost	Notes
CDERI transform `get_cderi` (if `lov` not passed)	$\mathcal{O}(N_{\text{aux}}\, n_{\text{ao}}^2\, o)$	N^4; repeated by every caller that does not pass `lov`
Build S_L, S_R	$\mathcal{O}(N_{\text{aux}} ov)$	two to three $N_{\text{aux}} \times ov$ arrays held at once
Grid fit, both integrals	$\mathcal{O}(N_{\text{it}} N_p ov)$	cheap, but runs every call
Main integral: $Q(\omega)$	$N_p\, N_{\text{aux}}^2 ov$	**dominant for small** n_t
Main integral: inv$(I+Q)$	$N_p\, N_{\text{aux}}^3$	explicit inverse
Main integral: target contractions	$N_p(2n_t N_{\text{aux}} ov + n_t N_{\text{aux}}^2)$	**dominant for large** n_t
Offset integral (Gauss–Laguerre)	$N_p \cdot 2n_t N_{\text{aux}} ov$	a second pass, as expensive as the target part of the main integral
$(A+B)^{-1}$	$N_{\text{aux}}^2 ov + N_{\text{aux}}^3 + 2n_t N_{\text{aux}} ov$	needed only because of the $(MP)^{1/2}(A+B)^{-1}$ form
Spin stacking (spin-orbital targets)	$\mathcal{O}(n_t^2 ov + n_t^3)$	`compress_target_rot` forms an $n_t \times n_t$ Gram
Higher moments	$2n_t N_{\text{aux}} ov$ per order	fine

The leading terms are therefore

$$\text{cost}(\eta_0) \approx 48\, N_{\text{aux}}^2 ov + {\sim}190\, n_t N_{\text{aux}} ov + 48\, N_{\text{aux}}^3, \qquad \text{memory} \approx (2\text{–}3)\, N_{\text{aux}} ov + (2 + n_{\text{mom}})\, n_t ov. \tag{8}$$

This is $\mathcal{O}(N^4)$ when $n_t = \mathcal{O}(N)$, and $\mathcal{O}(N^5)$ for the full moment ($n_t = ov$, which the code warns about). The energy costs ${\sim}96\, N_{\text{aux}}^2 ov$ (Q plus the back contraction at every point). For the xc-kernel path, replace N_{aux} by the 2–3× larger product rank and ov by $2ov$.

How the call pattern turns this into total cost.

- **RPA bath:** a new `ssRIdRRPA` is built for every fragment. So the fragment-independent $48\, N_{\text{aux}}^2 ov$ work (the $Q(\omega)$ matrices and inverses) and the grid fit are repeated N_{frag} times. With a dense Kronecker target of $n_t \sim o\, v_{\text{clus}}$ rows, each fragment is $\mathcal{O}(N^4)$, and the calculation is $\mathcal{O}(N^5)$ overall.
- **Screening:** one call with all fragments stacked. It forms the full $n_t \times 2ov$ result and then keeps only the diagonal fragment blocks $T_f \eta_0 T_f^{\mathsf{T}}$, so most of the $n_t \times ov$ back-contraction is thrown away.
- **Accuracy:** the error is controlled only by the nested-Clenshaw–Curtis estimate and a warning above `err_tol`$= 10^{-6}$ per element. It is not certified. Section 3.1 gives `momentGW`'s measurement of this same route at about 10^{-9} relative error on a small-gap system, and an unfitted 48-point Clenshaw–Curtis rule measured 5×10^{-7} on a larger one.

3 Bringing in the monomial / Davidson improvements

3.1 What those projects established

Both `monomial_GW` (a momentGW fork whose legacy Clenshaw–Curtis η_0 route *is* this Vayesta route, Gauss–Laguerre offset included) and `Davidson-gw` (`gwfast`) replaced the frequency integral with the **Hale–Higham–Trefethen (HHT) / Zolotarev rational approximation of the inverse square root**. They write it in the symmetric, $(A+B)^{-1}$-free form

$$\eta_0 = D^{1/2}\, \tilde{M}^{-1/2}\, D^{1/2}, \qquad \tilde{M} = D^2 + 4WW^{\mathsf{T}}, \qquad W = D^{1/2} L^{\mathsf{T}} \quad \text{(symmetric positive definite)}, \tag{9}$$

$$x^{-1/2} \approx \sum_{j=1}^{N_p} \frac{w_j}{x + s_j} \qquad \text{on a certified } [\lambda_{\min}, \lambda_{\max}] \supseteq \operatorname{spec}(\tilde{M}). \tag{10}$$

Each pole is one resolvent, reduced exactly to auxiliary space by Woodbury:

$$F_j = D\,(D^2 + s_j)^{-1} \quad \text{(diagonal)}, \tag{11}$$

$$C_j = I + 4\,LF_jL^{\mathsf{T}} \quad (N_{\text{aux}} \times N_{\text{aux}},\ \text{SPD}) \qquad \leftarrow \text{ one Gram, } N_{\text{aux}}^2 ov, \tag{12}$$

$$\eta_0 \approx \sum_j w_j \Big[F_j - 4\,F_jL^{\mathsf{T}}C_j^{-1}LF_j\Big]. \tag{13}$$

The components, all implemented and tested in at least one of the two repositories:

1. **Certified spectral interval.**

   $$\lambda_{\min} = \min(D)^2, \qquad \lambda_{\max} = \max(D^2) + 4\min\left(\|W\|_F^2,\ \|W\|_1\|W\|_\infty\right).$$

   `gwfast` also has a tighter `"spectral"` option, which uses the top eigenvalue of one $N_{\text{aux}} \times N_{\text{aux}}$ Gram. Both ends are padded outwards.

2. **Pole count from a measured scalar error.** The rule is the smallest N_p for which $\max_x |1 - \sqrt{x}\,r(x)| \le \text{tol}$ over the interval. N_p grows only like $\log\kappa\,\log(1/\text{tol})$. The coefficients come from AGM and descending-Landen elliptic functions in extended precision, so $1 - k^2$ is never formed (`momentGW/eta0.py`, `gwfast/quadrature.py`).

3. **No grid fit and no offset integral.** The rational sum absorbs the bare and first-order terms: at zero coupling each solve is the identity. This removes Vayesta's `"HO"`/Gauss–Laguerre second pass and its a-optimisation.

4. **Cholesky solves with residual checks instead of `inv`.** `momentGW` measured that this is an accuracy rule rather than a speed rule, because both are dominated by the $N_{\text{aux}}^2 ov$ apply.

5. **No $ov \times ov$ intermediate.** The code records intermediate shapes and a test asserts that none is formed.

6. **Natural auxiliary functions** (`gwfast.integral.natural_auxiliary_functions`). These rotate the CDERIs to the eigenvectors of LL^{T} above a threshold. The rotation is symmetric (`eigh`) and does not change the result above that threshold. Compare Vayesta's off-by-default, non-symmetric `eig` compression.

7. **Square root and inverse square root together** (`gwfast.rpa.get_root_operators`). The expansion

   $$\tilde{M}^{1/2} \approx \sum_j w_j [I - s_j(\tilde{M} + s_j)^{-1}]$$

   reuses the same C_j, so one pass over the poles gives both.

Measured results, quoted from those projects:

Source	Comparison	Result
`monomial_GW/ROADMAP.md`, Milestone 2	ozone/PBE (condition 6.7×10^4) η_0 vs a dense oracle	Clenshaw–Curtis-48: 1.15×10^{-9}; HHT-26: 2.1×10^{-13}
same	benzene/cc-pVDZ η_0 stage	20 poles vs 48 points, **1.82**× faster
`Davidson-gw/docs/hht-quadrature-benchmark-2026-09-22.md`	C_8H_{18}/def2-TZVPP ($ov = 15411$, $N_{\text{aux}} = 1148$)	unfitted Clenshaw–Curtis-48: 5×10^{-7}; HHT-20: 3×10^{-14}; **1.95**× faster at 20 poles, 2.72× at 16; peak memory roughly halved

3.2 Applying it to Vayesta's projected moments

Vayesta needs $T\eta_0$ for a general target T ($n_t \times ov$), not just $\eta_0 L^\mathsf{T}$ as in GW. The Woodbury form still applies:

$$T\eta_0 \approx \underbrace{T\ \mathrm{diag}\Big(\sum_j w_j F_j\Big)}_{\approx T} - 4\sum_j w_j \underbrace{(TF_jL^\mathsf{T})}_{n_t N_{\mathrm{aux}} ov}\ \underbrace{C_j^{-1}}_{\text{solve}}\ \underbrace{(LF_j)}_{n_t N_{\mathrm{aux}} ov}. \tag{14}$$

Per pole this costs $N_{\mathrm{aux}}^2 ov$ (Gram) + $N_{\mathrm{aux}}^3/3$ (Cholesky) + $2n_t N_{\mathrm{aux}} ov + n_t N_{\mathrm{aux}}^2$. The scaling class is the same as today. The gain comes from having fewer passes and dropping the redundant ones:

	Current (CC-48, `"HO"`)	HHT, $N_p \approx 20$	Ratio
$N_{\mathrm{aux}}^2 ov$ work	48	20	2.4×
$n_t N_{\mathrm{aux}} ov$ work	∼96 (main) +∼96 (offset) +2	∼40	∼4.8×
N_{aux}^3	48 explicit inverses + SVD	20 Cholesky factorisations	∼7×
Grid fitting, $(A+B)^{-1}$	yes	not needed	–
Error control	heuristic, about 10^{-7} to 10^{-9}	certified to tol (10^{-14} default in `momentGW`)	–

The expected η_0 speed-up is therefore about **2.4× when** $n_t \ll N_{\mathrm{aux}}$**, about 4× when** $n_t \sim N_{\mathrm{aux}}$**, and about 4.8× when** $n_t \gg N_{\mathrm{aux}}$, with accuracy improving by 5 to 7 orders of magnitude. These are operation-count estimates and must be measured. The one directly comparable measurement is 1.82× on the GW η_0 stage, where $n_t = N_{\mathrm{aux}}$.
The same pole factors give further savings that depend on what each caller actually needs:

1. **Cache the per-pole factors across targets and fragments.** F_j, C_j and its Cholesky factor depend only on the mean field, not on T. Build them once per calculation, at $N_p N_{\mathrm{aux}}^2 ov$ and $N_p N_{\mathrm{aux}}^2$ memory, and pass them to every caller. For the **RPA bath** this changes the fragment-independent part from $N_{\mathrm{frag}} \cdot 48\, N_{\mathrm{aux}}^2 ov$ to $20\, N_{\mathrm{aux}}^2 ov$ once. That is a reduction of $\sim 2.4\, N_{\mathrm{frag}}\times$ on that term, and it changes the overall scaling from $\mathcal{O}(N^5)$ to $\mathcal{O}(N^4)$ plus a per-fragment target term.

2. **Two-sided projections.** Screening keeps only $T_f \eta_0 T_f^\mathsf{T}$, and the RPA bath uses $m_0 T^\mathsf{T}$. Both are given by

$$T\eta_0 T^\mathsf{T} \approx TT^\mathsf{T} - 4\sum_j w_j\, B_j C_j^{-1} B_j^\mathsf{T}, \qquad B_j = TF_jL^\mathsf{T} \quad (n_t \times N_{\mathrm{aux}}). \tag{15}$$

   This never forms the $n_t \times ov$ result. Per pole and fragment the cost is $n_f N_{\mathrm{aux}} ov + n_f N_{\mathrm{aux}}^2 + n_f^2 N_{\mathrm{aux}}$, instead of $2n_t N_{\mathrm{aux}} ov$ for the stacked target followed by the projection. For screening it also stops computing the off-diagonal fragment blocks, which are discarded today.

3. **Kronecker-structured targets.** The bath target is $R_o \otimes R_v$, but it is stored densely, and `screening_moment` does the same per spin. Applying it in factored form, $B_j[I, A, Q] = \sum_{ia} R_o[I, i]\, R_v[A, a]\, (F_j L^\mathsf{T})[ia, Q]$ done as two small GEMMs, saves a factor of about o (or v) per contraction. With item 1, the bath then costs about $\mathcal{O}(o\, v\, v_{\mathrm{clus}} N_{\mathrm{aux}})$ per fragment and pole, which is $\mathcal{O}(N^3)$ per fragment and $\mathcal{O}(N^4)$ overall.

4. **Targets inside the RI span.** For the cumulant energy in `qemb` the target is L itself, and TF_jL^T *is* the Gram, $(C_j - I)/4$. This is exactly the `momentGW` kernel, at $\sim 2N_{\mathrm{aux}}^2 ov$ per pole and no n_t term.

5. **Energy from the same poles.** Applying the rule to $\sqrt{x} = x \cdot x^{-1/2}$ for both $\tilde{M}$ and D^2, and differencing so that the leading terms cancel identically, gives

$$\mathrm{Tr}\,(\tilde{M}^{1/2}) - \mathrm{Tr}(D) \approx 4\sum_j w_j s_j\ \mathrm{Tr}\Big[C_j^{-1}\big(L\, D\, (D^2 + s_j)^{-2} L^\mathsf{T}\big)\Big]. \tag{16}$$

This costs $\sim 2N_{\mathrm{aux}}^2 ov$ per pole, against $\sim 96\, N_{\mathrm{aux}}^2 ov$ for `NITrRootMP` today: about 2.4× faster and certified. One caveat: the bound is relative per eigenvalue, so the absolute energy error is at most about $2\,\mathrm{tol}\,\mathrm{Tr}(D)$. Choose tol against the required E_c accuracy. At 10^{-14} this is well under a microhartree for any system Vayesta handles.

6. **Natural auxiliary functions.** Rotating L once to $N'_{\mathrm{aux}} \le N_{\mathrm{aux}}$ NAFs saves $(N_{\mathrm{aux}}/N'_{\mathrm{aux}})^2$ on every Gram and $N_{\mathrm{aux}}/N'_{\mathrm{aux}}$ on every target contraction. The saving depends on the system and threshold and has not been measured here. The rotation must use a scale-aware PSD criterion and report the rank and discarded norm, which is `monomial_GW` Milestone 4's open item. The current absolute 10^{-12} cut does neither.
7. **Symmetric gap and eigenvectors.** $\tilde{M}$ is symmetric with eigenvalues Ω^2. `get_gap` can therefore use the symmetric `pyscf.lib.davidson` on $x \mapsto D^2 x + 4W(W^{\mathsf{T}}x)$ ($\mathcal{O}(N_{\mathrm{aux}} ov)$ per matvec) instead of the left/right non-symmetric `eig` on MP followed by re-biorthogonalisation.

Rough projected totals for the two main embedding uses (operation counts, before NAF):

Use	Today	With the changes	Gain
RPA bath, N_{frag} fragments	$N_{\mathrm{frag}}\big(48N_{\mathrm{aux}}^2 ov + \sim 190\, o\, v_c N_{\mathrm{aux}} ov\big) \sim \mathcal{O}(N^5)$	$20N_{\mathrm{aux}}^2 ov + N_{\mathrm{frag}} \cdot 20\big(o\, v\, v_c N_{\mathrm{aux}} + \dots\big) \sim \mathcal{O}(N^4)$	grows with system size; already about $N_{\mathrm{frag}} \times o$ on the target term
`mrpa` screening	$48N_{\mathrm{aux}}^2 ov + \sim 190\, n_t N_{\mathrm{aux}} ov$, $n_t = \sum_f 2ov_f$	$20N_{\mathrm{aux}}^2 ov + 20\sum_f \big(2n_f N_{\mathrm{aux}} ov/k + n_f N_{\mathrm{aux}}^2\big)$, where k is the Kronecker saving	about 5×, and more with Kronecker targets
Total dRPA energy	$\sim 96 N_{\mathrm{aux}}^2 ov$	$\sim 40 N_{\mathrm{aux}}^2 ov$	about 2.4×, certified

3.3 Proposed sequence

This mirrors `monomial_GW`'s ordering: restricted dRPA first, measured before it becomes the default, and the old route kept as an independent reference.

1. **Baseline.** Record η_0, the projected moments, energies, screened interactions and bath occupations for a small set (H_2O, a small-gap molecule, one Hubbard case, one solid), using the existing `vayesta/tests/rpa/test_rirpa_*` systems plus a dense `ssRPA` oracle.
2. **HHT kernel for `ssRIdRRPA`.** Port `momentGW/eta0.py` (interval, poles, scalar certificate) and a projected apply that accepts T in one-sided, two-sided and in-RI-span forms. Add an `eta0_method={"clencur","hht"}` option. Keep Clenshaw–Curtis as the default until the baseline agrees within the certified error, then switch.
3. **Pole-factor cache and caller changes.** Build the factors once per embedding object. Move `bath/rpa.py` and `screening_moment.py` to the two-sided, Kronecker-factored apply, and stop recomputing CDERIs per fragment by passing `lov` everywhere.
4. **Energy and gap** from the same poles (items 5 and 7 above).
5. **NAF/scale-aware compression** behind an option, reporting rank, discarded norm and the observed effect on η_0 and energy.
6. **Deferred: the UHF and xc-kernel paths.** `ssRIURPA` should follow with the stacked-spin convention that `gwfast` already validated (spin factor 1 over $2ov$). EDMET's `rixc` path cannot use the symmetric $D^{1/2}\tilde{M}^{-1/2}D^{1/2}$ form, because $A - B$ is not diagonal. It needs either a non-symmetric Woodbury rational apply (LU solves) on MP, whose interval is not theorem-backed, or $(A-B)^{1/2}$ as a diagonal-plus-low-rank function. Keep it on Clenshaw–Curtis until that is designed. Periodic `kdf` cases also wait, for the reasons `monomial_GW` Milestone 6 records.

3.4 Points to settle before implementation

- **Spin conventions.** `ssRIdRRPA` mixes spatial integrals with spin-orbital outputs (`stack_spin`, `return_spatial`). The factor 4 in $\tilde{M}$ and the split of the spin-diagonal identity term must be checked against the dense `ssRPA` before anything else.
- **Negative CDERIs** (`lov_neg`, from non-PSD metrics in some periodic setups). $A + B$ then takes an indefinite low-rank part, C_j is no longer SPD, and the certified interval argument fails. The HHT path should refuse them at first.
- **Memory for the cache.** Storing the factors costs $N_p N_{\mathrm{aux}}^2$ (about 0.2 GB at $N_{\mathrm{aux}} = 1000$, $N_p = 20$). For very large N_{aux}, rebuild per pole; `gwfast`'s `PRECOMPUTE_S` switch is exactly this trade.
- **Higher moments** are already exact recurrences at $\mathcal{O}(n_t N_{\mathrm{aux}} ov)$ per order. `monomial_GW` Milestone 3 found that its high-order instabilities were in the downstream realisation, not in this recurrence, so no change is proposed there.